



Web-Based Application Development (CSE3023)

Lab Report 7

Group:

K1

Prepared by:

Wong Cai Yi S76084

Program:

Bachelor of Computer Science (Software Engineering)
with Honours

Date of Submission: 19 May 2026

Lab Exercise

subjectBean.java

The screenshot displays an IDE with a project structure on the left and a code editor on the right. The project structure shows a web application with various files and packages. The code editor shows the implementation of the `subjectBean` class.

Project Structure (Left Panel):

- Lab 7 [main]
 - Web Pages
 - META-INF
 - context.xml
 - WEB-INF
 - web.xml
 - subject
 - registerSubject.jsp [R/M]
 - updateSubject.jsp [-/A]
 - viewSubjects.jsp [-/A]
 - dashboard.jsp [-/M]
 - index.html
 - login.html
 - register.html
 - Remote Files
 - bootstrap.bundle.min.js
 - bootstrap.min.css
 - Source Packages
 - bean
 - StudentBean.java
 - subjectBean.java
 - controller
 - UserServlet.java
 - subjectServlet.java [-/M]
 - dao
 - StudentDAO.java
 - subjectDAO.java [-/M]
 - Test Packages
 - <default package>
 - Libraries
 - mysql-connector-j-9.6.0.jar
 - JDK 23 (Default)
 - Apache Tomcat or TomEE
 - Test Libraries
 - Apache Tomcat or TomEE (embeddable)
 - Configuration Files
 - MANIFEST.MF
 - context.xml

Code Editor (Right Panel):

```
1  /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to view the code.
3  * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit the template
4  */
5  package bean;
6
7  /**
8   *
9   * @author Asus
10  */
11  public class subjectBean {
12
13      private int id;
14      private String matric_no;
15      private String subject_code;
16      private String subject_name;
17
18      public int getId() {
19          return id;
20      }
21
22      public void setId(int id) {
23          this.id = id;
24      }
25
26      public String getMatric_no() {
27          return matric_no;
28      }
29
30      public void setMatric_no(String matric_no) {
31          this.matric_no = matric_no;
32      }
33
34      public String getSubject_code() {
35          return subject_code;
36      }
37
38      public void setSubject_code(String subject_code) {
39          this.subject_code = subject_code;
40      }
41
42      public String getSubject_name() {
43          return subject_name;
44      }
45
46      public void setSubject_name(String subject_name) {
47          this.subject_name = subject_name;
48      }
49
50  }
51
52  }
```

subjectServlet.java

```
1 package controller;
2
3 import bean.StudentBean;
4 import bean.subjectBean;
5 import dao.subjectDAO;
6
7 import java.io.IOException;
8 import java.util.List;
9 import javax.servlet.ServletException;
10
11 import javax.servlet.http.HttpServlet;
12 import javax.servlet.http.HttpServletRequest;
13 import javax.servlet.http.HttpServletResponse;
14 import javax.servlet.http.HttpSession;
15
16
17 public class subjectServlet extends HttpServlet {
18
19     private subjectDAO dao = new subjectDAO();
20
21     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
22         throws ServletException, IOException {
23         response.setContentType("text/html;charset=UTF-8");
24
25         // 1. Session Verification Guard
26         HttpSession session = request.getSession(false);
27         if (session == null || session.getAttribute("loggedUser") == null) {
28             response.sendRedirect(request.getContextPath() + "/login.html");
29             return;
30         }
31
32         // 2. Extract identity parameters implicitly from session
33         StudentBean loggedUser = (StudentBean) session.getAttribute("loggedUser");
34         String matricNo = loggedUser.getMatricNo();
35
36         // 3. Operational action routing parameter
37         String action = request.getParameter("action");
38
39         if (action == null) {
40             action = "list";
41         }
42
43         try {
44             switch (action) {
45                 case "insert":
46                     insertSubject(request, response, matricNo);
47                     break;
48                 case "delete":
49                     deleteSubject(request, response, matricNo);
50                     break;
51                 case "edit":
52                     showEditForm(request, response);
53                     break;
54                 case "update":
55                     updateSubject(request, response, matricNo);
56                     break;
57                 case "list":
58                     listSubjects(request, response, matricNo);
59                     break;
60                 default:
61                     listSubjects(request, response, matricNo);
62                     break;
63             }
64         } catch (Exception e) {
65             e.printStackTrace();
66             throw new ServletException(e);
67         }
68
69     private void listSubjects(HttpServletRequest request, HttpServletResponse response, String matricNo)
70         throws ServletException, IOException {
71         List<subjectBean> list = dao.getSubjectsByMatric(matricNo);
72         request.setAttribute("subjectList", list);
73         request.getRequestDispatcher("subject/viewSubjects.jsp").forward(request, response);
74     }
75 }
```

```

73
74
75     private void insertSubject(HttpServletRequest request, HttpServletResponse response, String matricNo)
76     {
77         throws IOException {
78             String code = request.getParameter("subjectCode");
79             String name = request.getParameter("subjectName");
80
81             subjectBean sub = new subjectBean();
82             sub.setMatric_no(matricNo);
83             sub.setSubject_code(code);
84             sub.setSubject_name(name);
85
86             dao.registerSubject(sub);
87             response.sendRedirect(request.getContextPath() + "/subjectServlet?action=list");
88         }
89
90     private void showEditForm(HttpServletRequest request, HttpServletResponse response)
91     {
92         throws ServletException, IOException {
93             int id = Integer.parseInt(request.getParameter("id"));
94             subjectBean existingSubject = dao.getSubjectById(id);
95             request.setAttribute("subject", existingSubject);
96             request.getRequestDispatcher("subject/updateSubject.jsp").forward(request, response);
97         }
98
99     private void updateSubject(HttpServletRequest request, HttpServletResponse response, String matricNo)
100     {
101         throws IOException {
102             // 1. Ensure the id hidden input can be read properly
103             int id = Integer.parseInt(request.getParameter("id"));
104
105             // 2. These parameter strings are case-sensitive!
106             // They MUST match the 'name' attribute inside your HTML input elements exactly.
107             String code = request.getParameter("subjectCode");
108             String name = request.getParameter("subjectName");
109
110             subjectBean sub = new subjectBean();
111             sub.setId(id);
112             sub.setMatric_no(matricNo);
113             sub.setSubject_code(code);
114             sub.setSubject_name(name);
115
116             dao.updateSubject(sub);
117
118             // Your dynamic context path redirect that we fixed earlier
119             response.sendRedirect(request.getContextPath() + "/subjectServlet?action=list");
120         }
121
122     private void deleteSubject(HttpServletRequest request, HttpServletResponse response, String matricNo)
123     {
124         throws IOException {
125             int id = Integer.parseInt(request.getParameter("id"));
126             dao.deleteSubject(id, matricNo);
127             response.sendRedirect(request.getContextPath() + "/subjectServlet?action=list");
128         }
129
130     @Override
131     protected void doGet(HttpServletRequest request, HttpServletResponse response)
132     {
133         throws ServletException, IOException {
134             processRequest(request, response);
135         }
136
137     @Override
138     protected void doPost(HttpServletRequest request, HttpServletResponse response)
139     {
140         throws ServletException, IOException {
141             processRequest(request, response);
142         }
143     }

```

subjectDAO.java

```
1 package dao;
2
3 import bean.subjectBean;
4 import java.sql.Connection;
5 import java.sql.DriverManager;
6 import java.sql.PreparedStatement;
7 import java.sql.ResultSet;
8 import java.util.ArrayList;
9 import java.util.List;
10
11 public class subjectDAO {
12
13     private Connection getConnection() throws Exception {
14         Class.forName("com.mysql.cj.jdbc.Driver");
15         // Ensure the password matches your local XAMPP setup ("root" with "" or "admin")
16         return DriverManager.getConnection("jdbc:mysql://localhost:3306/lab7_db", "root", "admin");
17     }
18
19     // CREATE
20     public boolean registerSubject(subjectBean subject) {
21         String sql = "INSERT INTO registered_subjects (matric_no, subject_code, subject_name) VALUES (?, ?, ?)";
22         try (Connection conn = getConnection();
23             PreparedStatement ps = conn.prepareStatement(sql)) {
24             ps.setString(1, subject.getMatric_no());
25             ps.setString(2, subject.getSubject_code());
26             ps.setString(3, subject.getSubject_name());
27             return ps.executeUpdate() > 0;
28         } catch (Exception e) {
29             e.printStackTrace();
30             return false;
31         }
32     }
33
34     // READ (Fetch items belonging exclusively to the active user session)
35     public List<subjectBean> getSubjectsByMatric(String matricNo) {
36         List<subjectBean> list = new ArrayList<>();
37         String sql = "SELECT * FROM registered_subjects WHERE matric no = ?";
38         try (Connection conn = getConnection();
39             PreparedStatement ps = conn.prepareStatement(sql)) {
40             ps.setString(1, matricNo);
41             try (ResultSet rs = ps.executeQuery()) {
42                 while (rs.next()) {
43                     subjectBean sub = new subjectBean();
44                     sub.setId(rs.getInt("id"));
45                     sub.setMatric_no(rs.getString("matric_no"));
46                     sub.setSubject_code(rs.getString("subject_code"));
47                     sub.setSubject_name(rs.getString("subject_name"));
48                     list.add(sub);
49                 }
50             }
51         } catch (Exception e) {
52             e.printStackTrace();
53         }
54         return list;
55     }
56
57     // READ SINGLE (For populating the update screen inputs)
58     public subjectBean getSubjectById(int id) {
59         String sql = "SELECT * FROM registered_subjects WHERE id = ?";
60         try (Connection conn = getConnection();
61             PreparedStatement ps = conn.prepareStatement(sql)) {
62             ps.setInt(1, id);
63             try (ResultSet rs = ps.executeQuery()) {
64                 if (rs.next()) {
65                     subjectBean sub = new subjectBean();
66                     sub.setId(rs.getInt("id"));
67                     sub.setMatric_no(rs.getString("matric_no"));
68                     sub.setSubject_code(rs.getString("subject_code"));
69                     sub.setSubject_name(rs.getString("subject_name"));
70                     return sub;
71                 }
72             }
73         }
74     }
75 }
```

```

73     } catch (Exception e) {
74         e.printStackTrace();
75     }
76     return null;
77 }
78
79 // UPDATE
80 // Locate this method in subjectDAO.java and ensure it matches this exactly:
81 public boolean updateSubject(subjectBean subject) {
82     // Crucial check: make sure your table columns are spelled exactly subject_name and subject_code
83     String sql = "UPDATE registered_subjects SET subject_name = ?, subject_code = ? WHERE id = ? AND matric_no = ?";
84     try (Connection conn = getConnection();
85         PreparedStatement ps = conn.prepareStatement(sql)) {
86
87         ps.setString(1, subject.getSubject_name());
88         ps.setString(2, subject.getSubject_code());
89         ps.setInt(3, subject.getId());
90         ps.setString(4, subject.getMatric_no());
91
92         return ps.executeUpdate() > 0;
93     } catch (Exception e) {
94         e.printStackTrace();
95         return false;
96     }
97 }
98
99 // DELETE
100 public boolean deleteSubject(int id, String matricNo) {
101     String sql = "DELETE FROM registered_subjects WHERE id = ? AND matric_no = ?";
102     try (Connection conn = getConnection();
103         PreparedStatement ps = conn.prepareStatement(sql)) {
104         ps.setInt(1, id);
105         ps.setString(2, matricNo);
106         return ps.executeUpdate() > 0;
107     } catch (Exception e) {
108         e.printStackTrace();
109         return false;
110     }
111 }
112 }

```

Login.html

```

1  <!DOCTYPE html>
2  <!--
3  Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4  Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5  -->
6  <html>
7      <head><title>Student Login</title></head>
8      <body>
9          <h2>Student Login</h2>
10         <form action="UserServlet" method="POST">
11             <input type="hidden" name="action" value="login">
12             Matric No: <input type="text" name="matricNo" required><br><br>
13             Password: <input type="password" name="password" required><br><br>
14             <input type="submit" value="Login">
15         </form>
16     </body>
17 </html>
18

```

Dashboard.jsp

```
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <%@page import="bean.StudentBean"%>
9 <!DOCTYPE html>
10 <html lang="en">
11 <head>
12 <meta charset="UTF-8">
13 <meta name="viewport" content="width=device-width, initial-scale=1">
14 <title>Student Dashboard</title>
15 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
16 <style>
17 /* Custom CSS for profile image size and border */
18 .profile-img {
19 width: 160px;
20 height: 160px;
21 object-fit: cover;
22 border: 5px solid #198754; /* Bootstrap success green border */
23 }
24 </style>
25 </head>
26 <body class="bg-light">
27 <%
28 // 1. Session Verification Security Check (Prevents unauthenticated access)
29 response.setContentType("text/html; charset=UTF-8");
30 StudentBean user = (StudentBean) session.getAttribute("loggedInUser");
31 if (user == null) {
32 // If no active session exists, kick the user back to the login screen
33 response.sendRedirect("login.html");
34 return;
35 }
36 %>
37
38 <div class="container d-flex justify-content-center align-items-center" style="min-height:100vh;">
39 <div class="card shadow-lg border-0 rounded-4 p-4 p-md-5 text-center" style="max-width: 600px; width: 100%;">
40 <div class="card-body">
41
42 
44
45 <h2 class="fw-bold text-dark mb-2">Welcome, <%= user.getFullname() %>!</h2>
46 <p class="text-muted fs-5 mb-4">
47 Matric Number: <strong class="text-secondary"><%= user.getMatricNo() %></strong>
48 </p>
49
50 <hr class="my-4">
51
52 <div class="d-flex justify-content-center gap-3 flex-wrap">
53
54 <a href="<%= request.getContextPath() %>/subjectServlet?action=list" class="btn btn-primary px-4 py-2 fw-semibold shadow
55 Manage My Subjects
56 </a>
57
58 <a href="UserServlet?action=logout" class="btn btn-dark px-4 py-2 fw-semibold">
59 Logout
60 </a>
61
62 <a href="UserServlet?action=delete"
63 onclick="return confirm('WARNING:\nAre you sure you want to delete your account permanently?\nThis action cannot be
64 class="btn btn-danger px-4 py-2 fw-semibold">
65 Delete Account
66 </a>
67
68 </div>
69
70 </div>
71 </div>
72
73 </div>
74
75 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
76 </body>
</html>
```

RegisterSubject.jsp

```
7 <%%page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html lang="en">
10 <head>
11 <title>Register New Subject</title>
12 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
13 </head>
14 <body class="bg-light p-5">
15 <div class="container bg-white p-4 shadow rounded" style="max-width: 500px;">
16 <h3 class="mb-4 text-center">Register New Subject</h3>
17 <form action=".." subjectServlet" method="POST">
18 <input type="hidden" name="action" value="insert">
19 <div class="mb-3">
20 <label class="form-label fw-semibold">Subject Code</label>
21 <input type="text" name="subjectCode" class="form-control" placeholder="e.g., CSM3023" required>
22 </div>
23 <div class="mb-3">
24 <label class="form-label fw-semibold">Subject Name</label>
25 <input type="text" name="subjectName" class="form-control" placeholder="e.g., Web App Development" required>
26 </div>
27 <button type="submit" class="btn btn-success w-100 py-2">Add Subject</button>
28 <a href=".." subjectServlet?action=list" class="btn btn-light w-100 mt-2 border">Back to Workspace</a>
29 </form>
30 </div>
31 </body>
32 </html>
33
```

updateSubject.jsp

```
7 <%%page contentType="text/html" pageEncoding="UTF-8"%>
8 <%%page import="bean.subjectBean"%>
9 <!DOCTYPE html>
10 <html>
11 <head>
12 <title>Edit Subject</title>
13 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
14 </head>
15 <body class="bg-light p-5">
16 <%
17     subjectBean sub = (subjectBean) request.getAttribute("subject");
18     if (sub == null) {
19         response.sendRedirect(request.getContextPath() + "/subjectServlet?action=list");
20         return;
21     }
22 %>
23 <div class="container bg-white p-4 shadow rounded" style="max-width: 500px;">
24 <h3 class="mb-4">Modify Course Details</h3>
25 <form action="${pageContext.request.contextPath}/subjectServlet" method="POST">
26 <input type="hidden" name="action" value="update">
27
28 <input type="hidden" name="id" value="<%= sub.getId() %>">
29
30 <div class="mb-3">
31 <label class="form-label fw-semibold">Subject Code</label>
32 <input type="text" name="subjectCode" class="form-control" value="<%= sub.getSubject_code() %>" required>
33 </div>
34
35 <div class="mb-3">
36 <label class="form-label fw-semibold">Subject Name</label>
37 <input type="text" name="subjectName" class="form-control" value="<%= sub.getSubject_name() %>" required>
38 </div>
39
40 <button type="submit" class="btn btn-warning w-100 py-2 text-dark fw-semibold">Save Changes</button>
41 <a href=".." subjectServlet?action=list" class="btn btn-light w-100 mt-2 border">Cancel</a>
42 </form>

```


viewSubjects.jsp

```
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <%@page import="java.util.List"%>
9 <%@page import="bean.subjectBean"%>
10 <!DOCTYPE html>
11 <html lang="en">
12 <head>
13 <title>My Registered Subjects</title>
14 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
15 </head>
16 <body class="bg-light p-5">
17 <div class="container bg-white p-4 shadow rounded">
18 <div class="d-flex justify-content-between align-items-center mb-4">
19 <h2>My Academic Workspace</h2>
20 <div>
21 <a href="dashboard.jsp" class="btn btn-secondary btn-sm">Main Dashboard</a>
22 <a href="subject/registerSubject.jsp" class="btn btn-success btn-sm">Register New Subject</a>
23 </div>
24 </div>
25 <table class="table table-bordered table-striped align-middle">
26 <thead class="table-dark">
27 <tr>
28 <th>Subject Code</th>
29 <th>Subject Name</th>
30 <th style="width: 200px;">Actions</th>
31 </tr>
32 </thead>
33 <tbody>
34 <%
35 List<subjectBean> list = (List<subjectBean>) request.getAttribute("subjectList");
36 if (list != null && !list.isEmpty()) {
37     for (subjectBean sub : list) {
38
39 <tr>
40 <td><%= sub.getSubject_code() %></td>
41 <td><%= sub.getSubject_name() %></td>
42 <td>
43 <a href="subjectServlet?action=edit&id=<%= sub.getId() %>" class="btn btn-warning btn-sm">Update</a>
44 <a href="subjectServlet?action=delete&id=<%= sub.getId() %>" class="btn btn-danger btn-sm"
45 onclick="return confirm('Are you sure you want to drop this subject?');">Delete</a>
46
47 </td>
48 </tr>
49 <%
50     } else {
51 <%
52 <tr>
53 <td colspan="3" class="text-center text-muted py-4">No subjects registered yet.</td>
54 </tr>
55 <% } %>
56 </tbody>
57 </table>
58 </div>
59 </body>
60 </html>
```

Output:

Login page

Student Login

Matric No:

Password:

Login

Dashboard



Welcome, WONG CAI YI!

Matric Number: S76084

 Manage My Subjects

Logout

Delete Account

Manage subject

My Academic Workspace

Main Dashboard

Register New Subject

Subject Code	Subject Name	Actions
CSE3025	Web Application	Update Delete

Register new Subject

Register New Subject

Subject Code

CSE3029

Subject Name

Web Design

Add Subject

Back to Workspace

My Academic Workspace

Main Dashboard

Register New Subject

Subject Code	Subject Name	Actions
CSE3025	Web Application	UpdateDelete
CSE3029	Web Design	UpdateDelete

Modify record

Modify Course Details

Subject Code

CSE3030

Subject Name

Web Design

Save Changes

Cancel

My Academic Workspace

Main Dashboard

Register New Subject

Subject Code	Subject Name	Actions
CSE3025	Web Application	UpdateDelete
CSE3030	Web Design	UpdateDelete

Delete Subject

The screenshot shows a web browser window with the address bar displaying `localhost:8080/Lab_7/subjectServlet?action=list`. The page title is "My Registered Subjects". A confirmation dialog box is open, displaying the text "localhost:8080 says" and "Are you sure you want to drop this subject?". The dialog has "OK" and "Cancel" buttons. The background page shows the "My Academic Workspace" header with "Main Dashboard" and "Register New Subject" buttons. Below the header is a table with two rows of subject data.

Subject Code	Subject Name	Actions
CSE3025	Web Application	Update Delete
CSE3030	Web Design	Update Delete

My Academic Workspace

		Main Dashboard	Register New Subject
Subject Code	Subject Name	Actions	
CSE3025	Web Application	Update	Delete